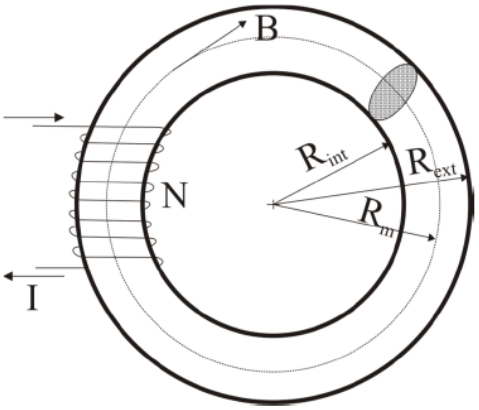
P7)



Directo: Con B o $\phi \rightarrow N.I$

$d_{ext} := 60cm$

$d_{int} := 40cm$

$B := 12000G$

Chapa aleada

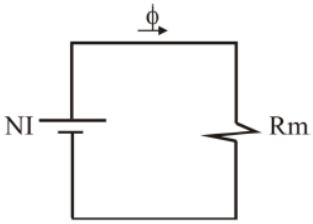
$d = \left(\frac{d_{ext} - d_{int}}{2} \right)$ $\quad r = \frac{d}{2}$ $\quad S = \pi \cdot r^2$ $\quad \boxed{S = 78{,}54\,cm^2}$

$B = \frac{\phi}{S}$ $\quad \phi = B \cdot S = 12.000 * 78{,}54 = 942.480$ [Mx]

$1Wb = 1 \times 10^8 \cdot Mx \Rightarrow \boxed{\phi = 9{,}425 \times 10^{-3} Wb}$ (a)

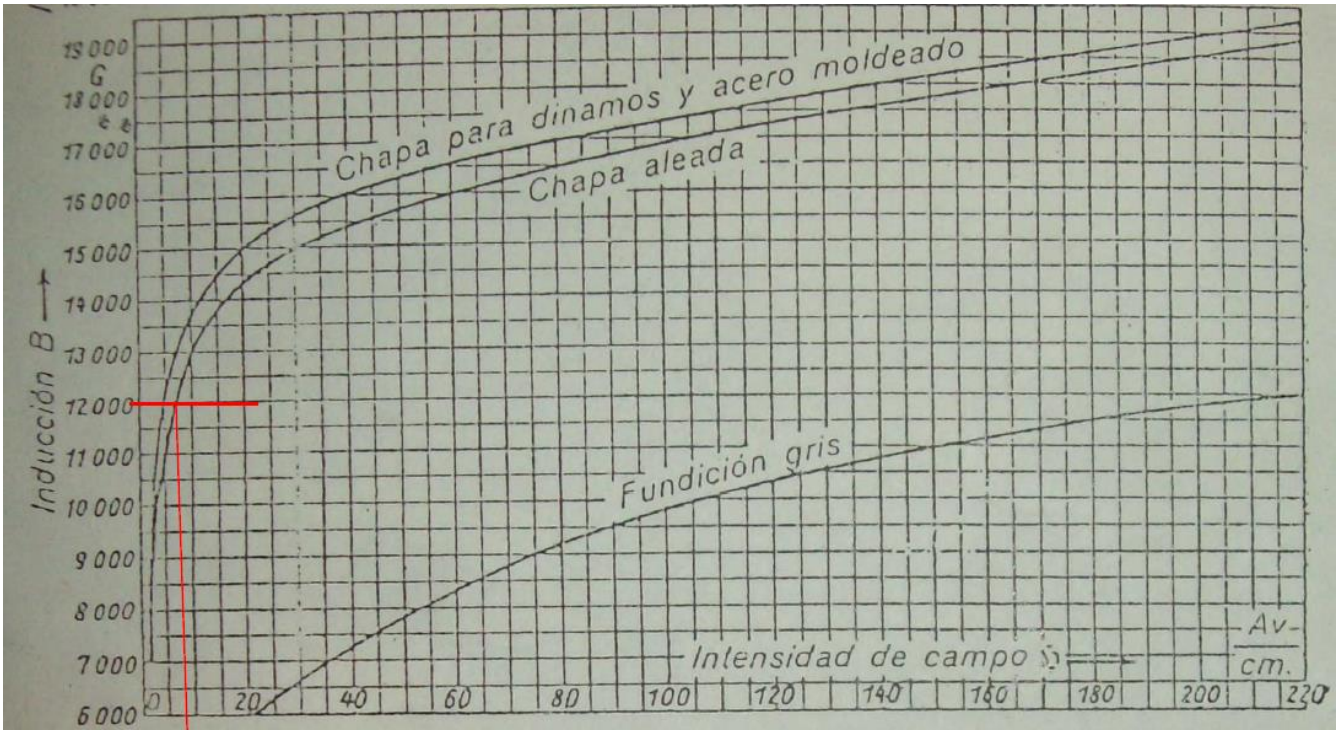
$d_m = \frac{(d_{ext} + d_{int})}{2}$ $\quad l_m = d_m \cdot \pi$

$d_m = 50 \cdot cm$ $\quad \boxed{l_m = 157{,}08\,cm}$



DE CURVA DE IMANTACIÓN

Entrando con B = 12.000 [G] se obtiene: $\boxed{H_{fe} = 7{,}5 \frac{A}{cm}}$



$\Sigma N.I = \Sigma \theta = \Sigma H.\ell$

$NI = Hfe * l_m = 7{,}5 * 157{,}08 = 1.178{,}1$ [Av] = θ: Fuerza Magneto Motriz o Excitación Magnética

b) $\mathcal{R}_m = \theta / \Phi = NI / \Phi = 1.178{,}1 / 9{,}425 * 10^{-3} = 124.997$ [Av/Wb] $\quad \boxed{\mathcal{R}_m = 124.997 \text{ [Av/Wb]}}$

c) Permeancia $\Lambda = 1 / \mathcal{R}_m = 1 / 124.997$ [Av/Wb]; $\boxed{\Lambda = 8 * 10^{-6} \text{ [Wb/Av]}}$

d) $I = \theta / N = 1.178 / 1.000 = 1{,}12$ [A] $\quad \boxed{I = 1{,}12 \text{ [A]}}$